CAS CS 538. Solutions to Problem Set 10

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How to submit: See instructions on Problem Set 1.

1 Commitments

Problem 1. [El Gamal commitments] We now consider a different commitment than the one from Discussion 10. Let $\mathbb{G}$ again be a group of prime order q generated by g. Consider the following protocol between Alice and Bob. Given a message $m \in \mathbb{G}$, Alice runs the following algorithm:

- <u>Commit(m)</u>: pick random $u \xleftarrow{R} \mathbb{G}$, $r \xleftarrow{R} \mathbb{Z}_q$ and let $c = (u, g^r, u^r \cdot m)$.

Alice sends c to Bob and keeps r to herself. Bob makes sure that he got three elements of $\mathbb{G}$.

Some time later, Bob wants to know what m is and Alice is ready to reveal it. Alice sends r and m to Bob. Of course, she may not be honest, so we will call what Bob receives m' and r'. Bob runs the following algorithm:

- <u>Verify(c, m', r')</u>: First, make sure that $m' \in \mathbb{G}$ and $r' \in \mathbb{Z}_q$. Let $c = (u, v, z)$. Check that $v = g^{r'}$ and that $z = u^{r'} m'$. If so, accept m' as the correct message. Else, reject.

Note that this commitment scheme is just the encryption scheme known as multiplicative ElGamal (Exercise 11.5), but without decryption.[1]

(a) Prove that this commitment scheme is *computationally hiding* under the DDH assumption.

The definition of computationally hiding commitments is essentially the same as the definition of semantically security encryption schemes: given $\mathbb{G}, g$, the adversary outputs m_0 and m_1, gets a commitment to m_b for random bit b and outputs $\hat{b}$, with the usual measure of adversarial advantage $\mathsf{Comadv} = |\Pr[\hat{b} = 1 \mid b = 0] - \Pr[\hat{b} = 1 \mid b = 1]|$.

Solution. This will be proven by reducing the game down to DDH.

Assume, for contradiction, that some eff. adv. $\mathcal{A}$ exists, that can crack the hiding property with non-negligible advantage ϵ. Note that $\mathcal{A}$'s advantage in the bit-guessing version of this game would be $\epsilon^* = \epsilon/2$, with $\epsilon^* = |\Pr[\hat{b} = b] - 1/2|$ (textbook and class, Boneh-Shoup: 2.2.5).

Now let's make up another adv, $\mathcal{B}$. $\mathcal{B}$ will be attacking $\mathcal{C}_{\text{DDH}}$, and will use $\mathcal{A}$ to do so:

- $\mathcal{B}$ will get $(u, v, w) \in \mathbb{G}^3$ from $\mathcal{C}_{\text{DDH}}$.
  - In Exp 0, $(u, v, w) = (g^\alpha, g^\beta, g^{\alpha\beta})$.

[1] In fact, a similar commitment scheme can be built out of any encryption scheme that uses randomness r: to commit to m, output pk and $E(pk, m)$ where E uses randomness r and no other randomness. To verify a revealed message m', use r' to re-encrypt m' under pk and check that the result matches the received commitment. It works as long as for every public key — even adversarially generated — correct decryption is unique.

- In Exp 1, $(u, v, w) = (g^\alpha, g^\beta, g^\gamma)$ for independent and rnd $\alpha, \beta, \gamma \in \mathbb{Z}_q$.

- $\mathcal{B}$ will run $\mathcal{A}$ on input $(\mathbb{G}, g)$. $\mathcal{A}$ will return $m_0, m_1 \in \mathbb{G}$.
- $\mathcal{B}$ will then choose a rnd bit $b \xleftarrow{R} \{0, 1\}$.
- $\mathcal{B}$ will then make the preliminary commitment $c = (u, v, w \cdot m_b)$ and then send it over to $\mathcal{A}$.
  - Note: Since $\mathbb{G}$ is a prime-order cyclic group generated by g, getting $u \xleftarrow{R} \mathbb{G}$ is the same thing as getting g^α for a rnd α. This matches the commitment's distribution.
- $\mathcal{A}$ returns $\hat{b}$. If $\hat{b} = b$, $\mathcal{B}$ outputs 1. Otherwise it'll output 0.

Analysis
Note, the main difference in the two experiments is in what $\mathcal{C}_{\text{DDH}}$ runs:

Exp 0 $\mathcal{C}_{\text{DDH}}$ will run $w = g^{\alpha\beta}$. Then, $c = (g^\alpha, g^\beta, (g^\alpha)^\beta \cdot m_b) = (u, g^r, u^r \cdot m_b)$ with $r = \beta$. This is just a valid commitment m_b. So, $\mathcal{A}$ is playing with the real hiding game. And by definition of bit-guessing probabilities:

- $\Pr[\mathcal{B} \to 1 \mid \text{Exp } 0] = \Pr[\hat{b} = b] = \frac{1}{2} + \epsilon^*$

Exp 1 Now, $\mathcal{C}_{\text{DDH}}$ is running $w = g^\gamma$ with it being uniform in $\mathbb{G}$. Note that, in this case, the third component $w \cdot m_b$ will also be uniformly distributed over $\mathbb{G}$ (since multiplication by a fixed group element is a bijection on $\mathbb{G}$). Consequently, c is independant of b. This means that $\mathcal{A}$ can do no better than just rnd guessing:

- $\Pr[\mathcal{B} \to 1 \mid \text{Exp } 1] = 1/2$

$\mathcal{B}$'s DDH advantage can then be calculcated as such:

$$\text{DDHAdv}[\mathcal{B}, \mathbb{G}] = |\Pr[\mathcal{B} \to 1 \mid \text{Exp } 0] - \Pr[\mathcal{B} \to 1 \mid \text{Exp } 1]| = \left| \left(\frac{1}{2} + \epsilon^* \right) - \frac{1}{2} \right| = \epsilon^* = \frac{\epsilon}{2}.$$

Since ϵ is non-negligible, $\text{DDHAdv}[\mathcal{B}, \mathbb{G}]$ is also non-negligible. This contradicts the DDH assumption, and therefore $\mathcal{A}$ can't exist. This proves how this commitment scheme is computationally hiding.

(b) Prove that the commitment is *perfectly binding*: namely, no matter how clever or computationally unbounded Alice is, for every c and every r_1, r_2, there is at most one m that Bob will accept. In other words, for every c, r_1, r_2, m_1, m_2, if $\text{Verify}(c, m_1, r_1)$ accepts and $\text{Verify}(c, m_2, r_2)$ also accepts, then $m_1 = m_2$.

Solution. This can be proven with algebraic properties (especially of prime-order groups).

Suppose an adv. makes a commitment $c = (u, v, z)$ and claims two valid openings $(m_1, r_1), (m_2, r_2)$. For this to actually work, this scheme will have to satisfy the following verification equations:

- For (m_1, r_1): $v = g^{r_1}$ and $z = u^{r_1} m_1$.
- For (m_2, r_2): $v = g^{r_2}$ and $z = u^{r_2} m_2$.

Uniqueness of r: Note the first components, $g^{r_1} = v = g^{r_2}$. Since $\mathbb{G}$ is a cyclic group of prime order q generated by g, the mapping of $r \mapsto g^r$ is a bijection from $\mathbb{Z}_q$ to $\mathbb{G}$. Then, $g^{r_1} = g^{r_2} \implies r_1 \equiv r_2 \pmod{q}$. And because both $r_1, r_2 \in \mathbb{Z}_q$, we must then have $r_1 = r_2$.

Uniqueness of m: Now we can sub $r_1 = r_2$ into the second components:

$$z = u^{r_1} m_1 \quad \text{and} \quad z = u^{r_2} m_2 = u^{r_1} m_2.$$

Equating the two expressions for z gives $u^{r_1} m_1 = u^{r_1} m_2$. Since $\mathbb{G}$ is a group, every element has a unique inverse. Multiplying both sides by $(u^{r_1})^{-1}$ will cancel the term, resulting in $m_1 = m_2$.

This proves that any commitment, c, will have AT MOST one message, m, that will pass verification. This proves this scheme's property of perfectly binding.

Problem 2. (20 points) The above scheme is *perfectly* binding and *computationally* hiding. In discussion 10 we covered Pedersen commitments, which are *computationally* biding and *perfectly* hiding.

To recap the discussion, a commitment scheme is *perfectly hiding* when for every g, h, m_0 and m_1, the distribution of outputs produced by Commit(m_0) is identical to the distribution of outputs produced by Commit(m_1). (Note that this protects Alice during the commit phase because Bob learns nothing about m, regardless of Bob's computational power or any computational assumptions.)

Prove that it's not possible to have a commitment scheme to be both perfectly binding and perfectly hiding.

Solution. This proof will touch into Shannon's impossibility theorem for perfect secrecy (and more broadly, the Pigeonhole Theorem).

Suppose, for the sake of contradiction, that there exists a commitment scheme with msg space $\mathcal{M}$, rnd space $\mathcal{R}$, and commit space $\mathcal{C}$ that is BOTH perfectly hiding and perfectly binding. We also assume that $|\mathcal{M}| \geq 2$ (it'd be trivial otherwise). The Commit algorithm picks $r \xleftarrow{R} \mathcal{R}$ uniformly at rnd.

For any msg $m \in \mathcal{M}$ and commit $c \in \mathcal{C}$, let $N_m(c)$ be the num. of rnd-ness values $r \in \mathcal{R}$ such that Commit$(m; r) = c$. Since r is uniform, the probability of making c from m is $\Pr[\text{Commit}(m) = c] = \frac{N_m(c)}{|\mathcal{R}|}$.

Perfect Hiding: Perfect hiding requires that $\forall m_0, m_1 \in \mathcal{M}$ and $\forall c \in \mathcal{C}$, the distributions are the same:

$$\Pr[\text{Commit}(m_0) = c] = \Pr[\text{Commit}(m_1) = c] \implies \frac{N_{m_0}(c)}{|\mathcal{R}|} = \frac{N_{m_1}(c)}{|\mathcal{R}|} \implies N_{m_0}(c) = N_{m_1}(c).$$

Therefore, $N_m(c)$ is independent of m. Now, let $N(c)$ be this common value. If $N(c) > 0$, then for EVERY $m \in \mathcal{M}$, there exists AT LEAST one rnd-ness r that opens c to m.

Perfect Binding: Perfect binding requires that for any commitment c, there's AT MOST one msg, m, that Bob will accept. Basically, if $\text{Verify}(c, m_1, r_1)$ and $\text{Verify}(c, m_2, r_2)$ both accept, then it should be the case that $m_1 = m_2$. This then implies that for any fixed c, there can be valid openings for AT LEAST one distinct msg, m.

End: Now, consider any $c \in \mathcal{C}$ that is in the support of the commitment distribution (this c has to exist, as commitments are generated). For this c, we will have $N(c) > 0$:

- By the consequence of Perfect Hiding, $N(c) > 0$ implies that c can be validly opened to EVERY $m \in \mathcal{M}$.
- By the consequence of Perfect Binding, c can be validly opened to AT MOST one $m \in \mathcal{M}$.

Now, these two conditions can only work IFF $|\mathcal{M}| \leq 1$. However, this contradicts the initial assumption that the scheme supports AT LEAST two distinct msgs.

Therefore, no commitment scheme can be both perfectly hiding and perfectly binding when $|\mathcal{M}| \geq 2$.

2 Weak PRF to PRF

Recall that in the attack game for a PRF F defined over $(\mathcal{K}, \mathcal{X}, \mathcal{Y})$, the adversary $\mathcal{A}$ submits a sequence of queries to the challenger, and the challenger picks a random key k and answers each query x_i with $F(k, x_i)$ in Experiment 0, and it picks a random function f from the set of all functions from $\mathcal{X}$ to $\mathcal{Y}$ and answers each query x_i with $f(x_i)$ in Experiment 1. Note that the adversary has full control over the queries.

We can, however, define a weaker notion of a PRF, called a *weak PRF*, where the adversary is only allowed to make function queries on random inputs. More specifically, in the weak PRF game, whenever the adversary $\mathcal{A}$ makes a query, the *challenger* picks a random $x_i \in \mathcal{X}$ and answers the query with $(x_i, F(k, x_i))$ in Experiment 0, and with $(x_i, f(x_i))$ in Experiment 1. As expected, the adversary's advantage in this game, denoted $\text{wPRF}\mathsf{adv}[\mathcal{A}, F]$, is defined as the difference between the probability that $\mathcal{A}$ outputs 1 in Experiment 0 and the probability that $\mathcal{A}$ outputs 1 in Experiment 1; and the PRF F is said to be a weak PRF if $\text{wPRF}\mathsf{adv}[\mathcal{A}, F]$ is negligible for all efficient adversaries $\mathcal{A}$. (See definition 4.3 in Section 4.4.1 of the textbook if you want more explanations.)

Problem 3. (35 points) Problem 11.1 in the textbook. Correction to the problem: let q be the total number of queries that $\mathcal{A}$ makes. Then

$$\text{PRF}^{\text{ro}}\mathsf{adv}[\mathcal{A}, \mathcal{F}] \leq \text{wPRF}\mathsf{adv}[\mathcal{B}, \mathcal{F}']$$

should be instead

$$\text{PRF}^{\text{ro}}\mathsf{adv}[\mathcal{A}, \mathcal{F}] \leq \text{wPRF}\mathsf{adv}[\mathcal{B}, \mathcal{F}'] + \frac{q^2}{2 \cdot |\mathcal{X}|}$$

Interpretation: We know that $\mathcal{A}$ is polytime bounded and hence it can make only a polynomial number of queries. Thus, q^2 is negligible if $|\mathcal{X}|$ is exponentially large. So this weak-PRF-to-PRF conversion works for large $|\mathcal{X}|$ (something like 256-bit is reasonable) but not for small $\mathcal{X}$.

Approach: Let $\mathcal{A}$ be a PRF adversary against F. Because we are in the random oracle model, $\mathcal{A}$ gets to ask two types of queries:

- q_p PRF queries m_i (for $1 \leq i \leq q_p$) to the PRF (in Game 0) or random function (in Game 1)
- q_H random oracle (RO) queries $\hat{m}_i$ (for $1 \leq i \leq q_H$) to H

The total number of queries is $q = q_p + q_H$.

Let p_0 be the probability that $\mathcal{A}$ outputs 1 in experiment 0 of the PRF game (queries answered by F), and p_1 be the probability that $\mathcal{A}$ outputs 1 in experiment 1 of the PRG game(queries answered by a random f).

Let $\mathcal{C}$ be the following challenger: pick a random function $H : \mathcal{M} \to \mathcal{X}$ and a random function $f' : \mathcal{X} \to \mathcal{Y}$; answer PRF queries m_i with $f'(H(m_i))$ and RO queries $\hat{m}_i$ with $H(\hat{m}_i)$. Let $p_{\mathcal{C}}$ be the probability that $\mathcal{A}$ outputs 1 when interacting with $\mathcal{C}$.

In the text two parts, you will prove $|p_0 - p_c|$ and $|p_c - p_1|$ are both small, giving you that $|p_0 - p_1|$ is small.

(a) Prove that $|p_0 - p_c| \leq \text{wPRF}\mathsf{adv}[\mathcal{B}, \mathcal{F}']$ by building an adversary $\mathcal{B}$ that uses $\mathcal{A}$ and plays the wPRF game of F' with advantage $|p_0 - p_c|$.

Hint: $\mathcal{B}$ has to figure out how to consistently answer RO queries of $\mathcal{A}$. For example, if $\mathcal{A}$ asks $\mathcal{B}$ for $H(m)$, $\mathcal{A}$ may later ask $\mathcal{B}$ for $F'(H(m))$ (or $f'(H(m))$, depending on which game it's in). $\mathcal{B}$ should prepare for this eventuality, because it can't ask its wPRF challenger for a particular input. (This is called "preparing" or "programming" the random oracle.)

Solution. For this proof, let $\mathcal{B}$ be the adv. that plays against the wPRF challenger for F'. $\mathcal{B}$ will use $\mathcal{A}$ to do this.

$\mathcal{B}$ will hold two associative arrays to ensure consistency between H and the PRF outputs:

Map_H Maps msgs $m \in \mathcal{M}$ to inputs $x \in \mathcal{X}$ (basically simulating $H(m)$).

Map_F Maps $x \in \mathcal{X}$ to outputs $y \in \mathcal{Y}$ (basically caching the wPRF challenger's responses).

Now, $\mathcal{B}$ will run $\mathcal{A}$, and answer its queries as such:

RO Query $\hat{m}$
- If $\hat{m} \in \text{Domain}(\text{Map}_H)$, return $\text{Map}_H[\hat{m}]$.
- Otherwise, $\mathcal{B}$ will run a query to its wPRF challenger, getting a fresh rnd pair (x, y). Then, $\mathcal{B}$ will set $\text{Map}_H[\hat{m}] \leftarrow x$, $\text{Map}_F[x] \leftarrow y$, and then return x to $\mathcal{A}$.

PRF Query m
- If $m \in \text{Domain}(\text{Map}_H)$, then let $x = \text{Map}_H[m]$, and return $\text{Map}_F[x]$ to $\mathcal{A}$.
- Else, $\mathcal{B}$ will run a query to its wPRF challenger, getting (x, y). $\mathcal{B}$ will then set $\text{Map}_H[m] \leftarrow x$, $\text{Map}_F[x] \leftarrow y$, and then return y to $\mathcal{A}$.

When $\mathcal{A}$ halts and returns b', $\mathcal{B}$ will just return the same b'.

Analysis: Now note, because the wPRF challenger always supplies inputs x, that are uniformly rnd in $\mathcal{X}$, the values assigned to $H(m)$ are uniform in $\mathcal{X}$, which is just a true Rnd Oracle. Consistency between $H(m)$ and the composed PRF, therefore, is guaranteed by the dual maps.

- **Exp 0 (Real F'):** The challenger will give $(x, F'(k', x))$ for a rnd key k'. In this case, $\mathcal{A}$'s PRF queries will be answered with $F'(k', H(m))$. This is the same thing as the real PRF game for $\mathcal{F}(m) = F'(H(m))$. Thus:
  - $\Pr[\mathcal{B} \to 1 \mid \text{Exp } 0] = p_0$.
- **Exp 1 (Rnd Func):** The challenger now gives $(x, f(x))$ for a truly rnd function. In this case, $\mathcal{A}$'s PRF queries will be answered with $f(H(m))$. Now, this just the same thing as the behavior of Challenger $\mathcal{C}$ defined in the problem. Therefore:
  - $\Pr[\mathcal{B} \to 1 \mid \text{Exp } 1] = p_C$

By the definition of the wPRF advantage:

$$\text{wPRFadv}[\mathcal{B}, F'] = |\Pr[\mathcal{B} \to 1 \mid \text{Exp } 0] - \Pr[\mathcal{B} \to 1 \mid \text{Exp } 1]| = |p_0 - p_C|.$$

Therefore, $|p_0 - p_C| \leq \text{wPRFadv}[\mathcal{B}, F']$, proving the OG statement of this question.

(b) Prove that $|p_c - p_1| \leq \frac{q^2}{2 \cdot |\mathcal{X}|}$ by the difference lemma.

Solution. Now this proof is just an extension to the previous question. I'll be making 2 more hybrid games, and then defining its "Bad Event".

Game C (p_C): The challenger will lazily sample a rnd oracle $H : \mathcal{M} \to \mathcal{X}$ and a rnd func $f' : \mathcal{X} \to \mathcal{Y}$. Upon getting an RO query $\hat{m}$, it will return $H(\hat{m})$. Upon getting a PRF query m, it returns $f'(H(m))$.

Game 1 (p_1): The challenger will lazily sample $H : \mathcal{M} \to \mathcal{X}$ and a rnd func $F : \mathcal{M} \to \mathcal{Y}$. Upon getting an RO query $\hat{m}$, it returns $H(\hat{m})$. On a PRF query m, it returns $F(m)$.

Assume that both games operate on the same underlying probability space. In other words, the lazy sampling for H will use the exact same rnd space in both games. Now, let W_C be the event that $\mathcal{A}$ outputs 1 in Game C, and W_1 be the event that $\mathcal{A}$ outputs 1 in Game 1.

Now, $\Pr[W_C] = p_C$ and $\Pr[W_1] = p_1$.

Z**:** Let Z be the event that a collision in H occurs on any two distinct inputs, queried by $\mathcal{A}$. Since $\mathcal{A}$ makes AT MOST $q = q_p + q_H$ total queries, then there should be AT MOST $\binom{q}{2} < \frac{q^2}{2}$ distinct pairs of queried inputs.

Analysis: Consider the behavior of both games conditioned on $\neg Z$ occurring. If $\neg Z$ holds, then all queried inputs m should be mapped to DISTINCT values in $\mathcal{X}$.

- In Game 1, the PRF outputs are $F(m)$, which by lazy sampling, are independent and rnd values in $\mathcal{Y}$.
- In Game C, the PRF outputs are $f'(H(m))$. Because $H(m)$ are all distinct and f' is a rnd func, $f'(H(m))$ are also independent and rnd values in $\mathcal{Y}$.

Thus the adversary's view in Game C and Game 1 are the exact same. This implies that $W_C \wedge \neg Z$ occurs IFF $W_1 \wedge \neg Z$ occurs.

End: Therefore, by the difference lemma, we get:

$$|p_C - p_1| = |\Pr[W_C] - \Pr[W_1]| \leq \Pr[Z].$$

Yet, this remains to bound $\Pr[Z]$. By the union bound over all pairs of queries, the probability that any specific pair collides is $1/|\mathcal{X}|$. Therefore:

$$\Pr[Z] \leq \binom{q}{2}\frac{1}{|\mathcal{X}|} = \frac{q(q-1)}{2|\mathcal{X}|} < \frac{q^2}{2|\mathcal{X}|}.$$

Therefore, these inequalities yields the expected result (from the OG statement):

$$|p_C - p_1| \leq \frac{q^2}{2|\mathcal{X}|}.$$

$\square$